DG THERM

Service Manual

(Edition March 2022)

Diagnostic Grifols, S.A.
Passeig Fluvial 24
08150 Parets del Vallès
Barcelona, SPAIN

(this Service Manual starts on page 2)

This manual uses the following symbols to facilitate the location of specific sections.

DANGER!

Any kind of harmful situation for people that can be diminished using this indication or that it is convenient the QUALIFIED TECHNICIAN knows it.

WARNING!

Any kind of injure for people for whom protection systems are provided, but it is convenient the QUALIFIED TECHNICIAN knows it.

CAUTION!

Any kind of harmful situation for the equipment or other materials. This situation can be diminished using this indication or with other protections.

NOTE:

Used for clarification and complementary or emphasized information.

RESPONSIBLE AUTHORITY	Individual or group responsible for the use and maintenance of the equipment, and for ensuring that operators are adequately trained.
OPERATOR	Person operating the equipment for its intended purpose. The operator should have received training appropriate for this purpose.
QUALIFIED TECHNICIAN	Person responsible for the installation, reparation and special maintenance of the equipment who has received specific training for this purpose.
All restrictions referred to the RESPONSIBLE AUTHORITY are valid for the OPERATOR.	

This instrument is protected by international patents applicable to the entire machine or parts thereof.

The manufacturers reserve the right to take legal actions in order to protect their interests.

This manual should be used in conjunction with the Instructions for Use.

The information found in this document may be changed without prior notification.

Contents:

Changes from previous version:

Date	Change
03/2007	NA
08/2008	Technical service program section has been updated to include the effect of the regulation temperature offset in the averaged temperature shown in the control window.
10/2009	- New 'Adj. Temp.' command description (Section 4.6.4 and 6.2.4). - New log message 'Configuration change: Temperature adjusted' (Section 7.1). - References to 'Technical Specifications' and 'Safety Information' (Section 1.4 and 2). - Included fuse values (Section 1.3.1.1). - Updated images. - Minor typographic errors.
07/2011	- Added warning in chapters 2 and 6. - Added error E2 in Troubleshooting section. - Added continuity measurement after corrective actions. - Updated S0709V0 drawing. - Added S1247V0 drawing. - Updated PCA INCUBATOR schematic.
03/2022	- Document format updated. - Updated section *6.1 Troubleshooting*. Communication errors during firmware update added. Warning note for the electric continuity test added. - Updated section *6.2.1 Updating the firmware (firmware version ≥ 1.3.0 and DG Terminal ≥ v1.3.0).* Firmware update procedure updated for firmware version ≥ 1.3.0 and DG Terminal v1.3.0. - Updated section *8 Drawings*. Creo View Express section added. Mechanical drawings table updated. - Electronic schematics removed. - Minor text modifications.

1 INSTRUMENT GENERAL OVERVIEW

This manual is oriented to the **QUALIFIED TECHNICIAN** and contains the necessary information for a proper and secure manipulation of the instrument.

Read carefully all the information of this manual before starting the manipulation of the instrument. It is important to read the information concerning maintenance, correct use and liquids management.

NOTE:
All the operations described in this manual must be carried out only by a **QUALIFIED TECHNICIAN**.

1.1 Overview

DG Therm incubator has been especially designed to carry out the thermostatisation of 24 gel cards and/or the corresponding sample tubes of those techniques that require incubation at 37ºC. For further information, consult the *Instructions for Use* of the gel cards for the different techniques.

Incubation will take place by means of a dry hot bath at 37ºC for a pre-established time of 15 minutes. The *DG Therm* incubator has two independent timers that will enable carrying out two independent incubation batches.

1.2 Parts

The *DG Therm* has four main parts:

- *Thermostatic block and heater*: *DG Therm* uses a silicone rubber heater glued to an aluminum block, which transmits the heat to the card.
- *Electronics:* includes the main PCA, the display and the keypad, described below.
- *Card elevator system*: when lids are opened, cards are lifted using a tray located in each incubation zone, which makes the card handling easier.
- *Enclosure*: includes the case and two lids that cover the card zone.

All parts of the *DG Therm* are shown in drawing S0798V0.

1.3 Electronics

1.3.1 PCA Incubator

The PCA Incubator contains the *DG Therm* circuits, responsible for the temperature regulation and the display and keypad control. The PCA contains the following:

- Digital circuits: The PCA has a Texas MSP430F149 micro-controller *(U4)* that controls the Triac by means of two temperature sensors *(U6, U7)* to regulate the temperature of the thermostatic block. The PCA also controls acoustic indicators, display panel and keypad signals. It has a non-volatile memory *(U8)* to store the configuration parameters and a RS232 interface *(U3)* for technical service tasks.

- Power: Includes an input voltage detector *(OPTO2)* to switch the control parameters from 110V to 220V, a Triac *(OPTO1)* connected to the AC input to control the heater, a thermal fuse *(F4)* to avoid overheating and the main supply *(U1)* to power the digital circuits.

- Fuses: There are 4 fuses in the PCA Incubator:

  - FUS1: Antisurge fuse 2A/250V
  - FUS2: Antisurge fuse 2A/250V
  - FUS3: Quick Blow fuse 250mA/250V
  - F4: Thermal and current fuse (80ºC/2.2A)

WARNING!

The main PCA includes High Voltage parts. Disconnect the mains cable before any operation with the PCA.

1.3.2 PCA Display

The *DG Therm* uses a custom display to show timer, temperature information and the incidences that could occur during the operation. The PCA Display contains a LCD with a backlight, a LCD Driver *(U3)*, a real time clock *(U2)* and a buzzer *(LS1)*. A two button keypad is connected to the PCA Display.

1.4 Technical Specifications

See chapter 3 *"Technical Specifications"* in the *"DG Therm – Instructions for Use"*.

2 Safety Information

Safety Information related to the *DG Therm* is included in chapter 2 *"Safety Information"* in the *"DG Therm – Instructions for Use"*.

WARNING!

To completely disconnect the apparatus from mains remove the mains plug from the socket outlet or the appliance coupler.

3 Installation of DG Therm

3.1 Unpacking

See section 5.3 *"Unpacking the equipment"* in the *"DG Therm – Instructions for Use"*.

3.2 Set-up

See chapter 5 *"Installation"* in the *"DG Therm – Instructions for Use"*.

3.3 IQ

See section 5.1 *"Installation requirements"* in the *"DG Therm – Instructions for Use"*.

3.4 OQ

See chapter 8 *"Operation qualification procedure (OQ)"* in the *"DG Therm – Instructions for Use"*.

3.5 Uninstalling – Shipping

See chapter 9 *"Transport and storage"* in the *"DG Therm – Instructions for Use"*.

3.6 Decontamination

See section 7.1.2 *"Decontamination of the equipment"* in the *"DG Therm – Instructions for Use"*.

3.7 Storage

See chapter 9 *"Transport and storage"* in the *"DG Therm – Instructions for Use"*.

3.8 Disposal

With the equipment properly decontaminated, follow the drawing S0709V0 (*DG Therm* Assembly) in the reverse order to disassembly all the parts.

Dispose of the different parts depending on the kind of material:

- Base: stainless steel.
- Screws: stainless steel.
- Heating block:
    - Remove the thermal blanket: electronic waste.
    - Remove the lids support with a tool: stainless steel.
    - Remove the heating block sealing joint: rubber.
    - Remove the thermal silicon: silicon.
    - Heating block: aluminum.
- PCA Incubator (main PCA): electronic waste.
- Grounding plate: stainless steel.
- PCA Display:
    - Remove the battery (yellow box): batteries (ion lithium).
    - Display holders: stainless steel.
    - PCA Display: electronic waste.
- Keypad: electronic waste.
- Lids: plastic PC V0.
- Case: plastic ABS+PC V0
- Display window: plastic PC V0.
- Trays: plastic PS V0

4 *DG Terminal* software

The technical service software, *DG Terminal*, allows the user to access to low-level functions of *DG Therm*, such as checking the individual operation of different parts of the machine, install new firmware versions, set the date and time of the real time clock or retrieve the life time of the machine, among others.

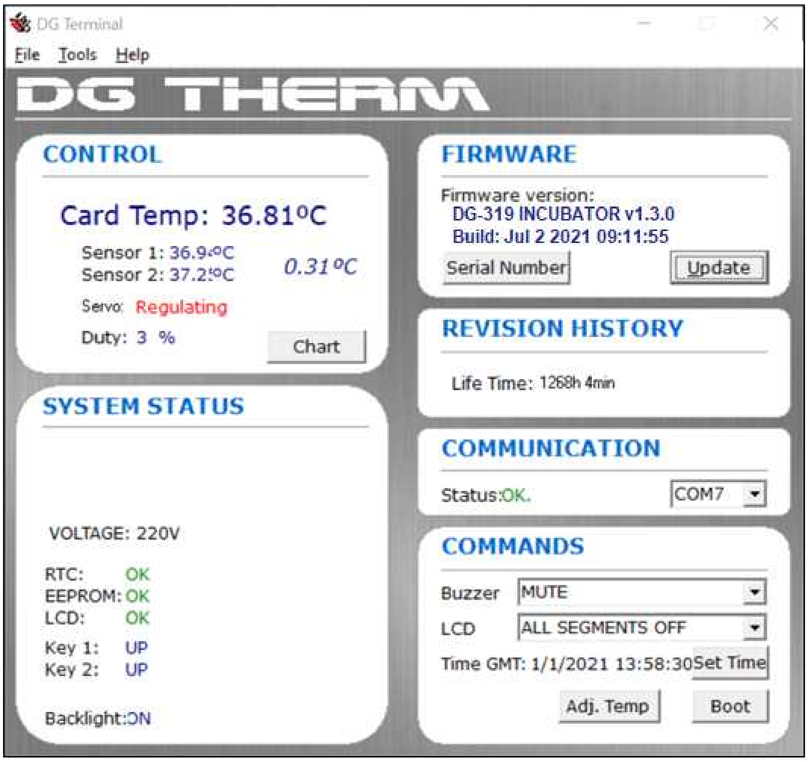


Figure 1. *DG Terminal* software main screen

4.1 CONTROL window

Shows all useful information about the temperature control system:

- Card Temp: Estimated temperature (ºC) inside the card calculated by means of the average temperature between the two sensors and the offset applied in the temperature regulation. Such offset modifies the temperature of the incubator to get the programmed temperature in the card and can be modified with a temperature adjustment procedure (see 6.2.4 Temperature adjustment).
- Temperature (ºC) measured by each temperature sensor.
- Deviation (ºC) between the two sensors.
- State of the servo control: warming up/regulating/stopped.
- Duty (%power) applied to the heater.

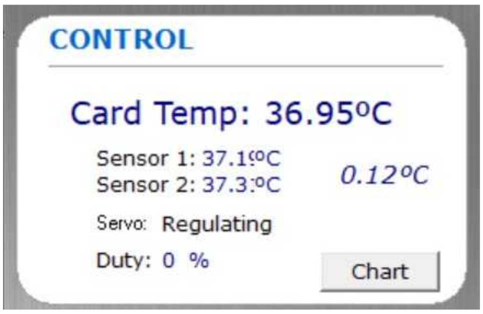


Figure 2. CONTROL window

When the *Chart* button is selected, a graph is shown with the applied power and the measured temperature versus time in seconds.

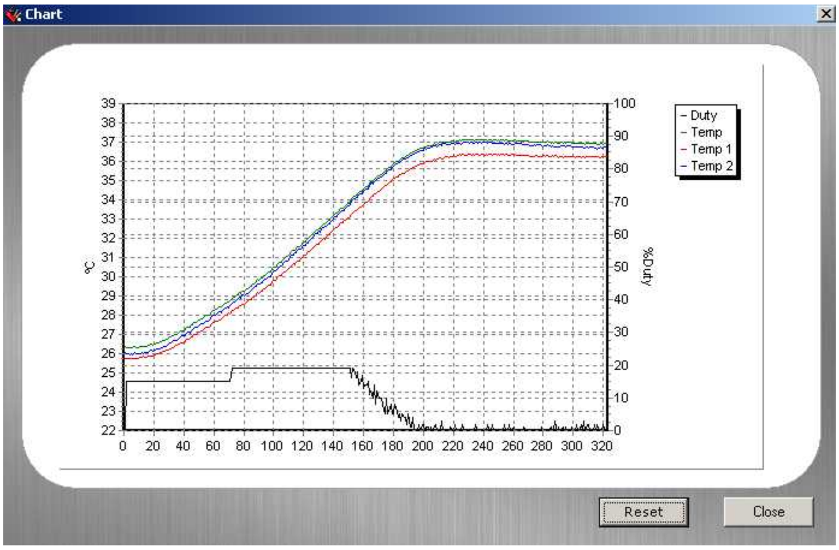


Figure 3. Power and measured temperature versus time

When the *DG Therm* is switched on and the *DG Terminal* is executed, the file ‘RegInc.txt’ is created in the same folder where the *DG Terminal* software is located. The structure of each line of this file is:

Card Temp [ºC]; Temp1(sensor1) [ºC]; Temp2(sensor2) [ºC]; Duty [%]

‘RegInc.txt’ is updated every second and there is not any limit for the size of the file. This file is reset pressing the *Reset* button located in the Chart window (Figure 3) or each time the DG Terminal is executed.

4.2 SYSTEM STATUS window

Shows the status of different parts of the incubator.

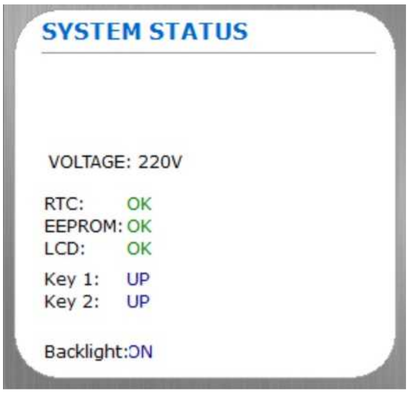


Figure 4. SYSTEM STATUS window

In case of the device mal functioning, the origin of the incidence will be reported in this window.

4.3 FIRMWARE window

Shows the actual firmware version installed into the microcontroller flash memory.

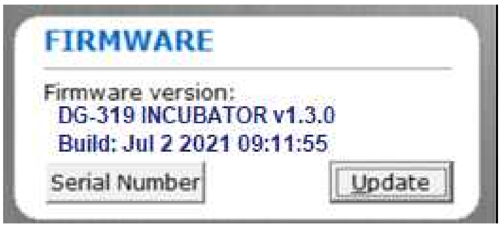


Figure 5. FIRMWARE window

4.3.1 Update Firmware command

Use the Update Firmware command from the File menu or the Update button from the FIRMWARE window to install a new firmware version into the microcontroller flash memory.

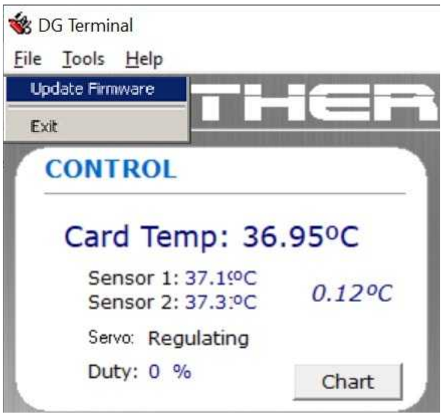


Figure 6. Update Firmware command in File menu

4.3.2 Serial Number command

Use the Serial Number button to enter and store the serial number of the device into its non volatile memory.

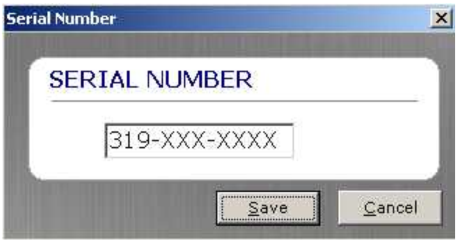


Figure 7. Serial Number window

When no serial number is yet saved into the non volatile memory, the number shown on the screen will be 319-XXX-XXXX.

The serial number is entered at the beginning of the manufacturing process and it needs to be reentered when replacing the PCA Incubator.

4.4 REVISION HISTORY window

Shows the life time (functioning hours) of the machine in hours and minutes.

Figure 8. REVISION HISTORY window

4.5 COMMUNICATION window

Shows the serial communication status of the device. Select the appropriate COM port if the status is not ok.

Figure 9. COMMUNICATION window

4.6 COMMANDS window

Within the commands window several sub systems of the device can be adjusted or checked.

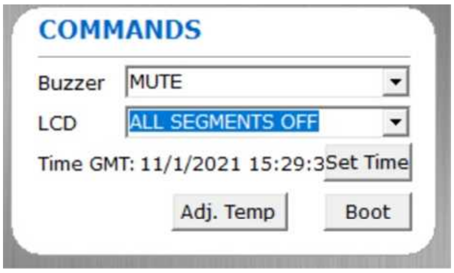


Figure 10. COMMANDS window

4.6.1 Buzzer command

For testing the functionality of the buzzer, different buzzer tones can be selected.

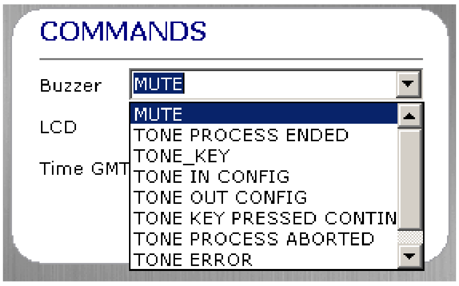


Figure 11. Buzzer command options

4.6.2 LCD command

For testing the functionality of the display, all segments of the display can be switched on or off.

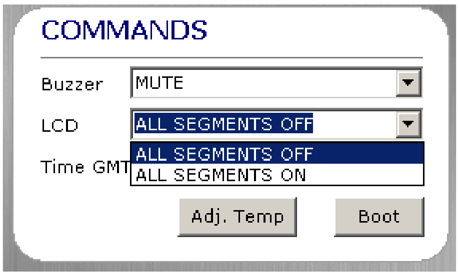


Figure 12. LCD command options

4.6.3 Set Time command

For adjusting the date and time of the real time clock. Time is stored in the device in GMT format (shown in the main window) and the *DG Terminal* converts from local time to update it.

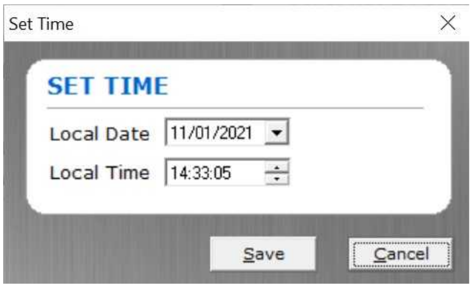


Figure 13. Set time window

4.6.4 Adjust Temperature command

Send the measured temperature inside the card lodgings with an external thermometer to the firmware. The incubator will calculate an internal offset to compensate the losses in the heat transmission (see 6.2.4. Temperature adjustment). This option is enabled from *DG Terminal* version ≥ 1.02 and with the firmware version ≥ 1.1.

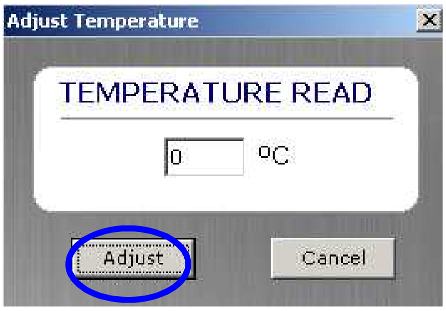


Figure 14. Adjust Temperature window

4.6.5 Boot command

Resets the device.

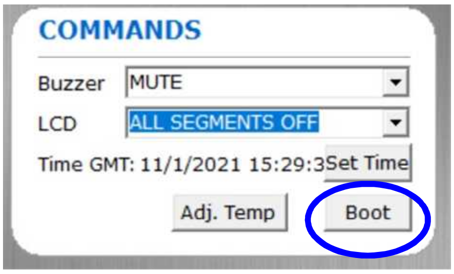


Figure 15. Boot command location

4.7 Tools menu options

All events during the device operation are stored into the non volatile memory.

To retrieve the stored event information, select within the Tools menu, the Log viewer option.

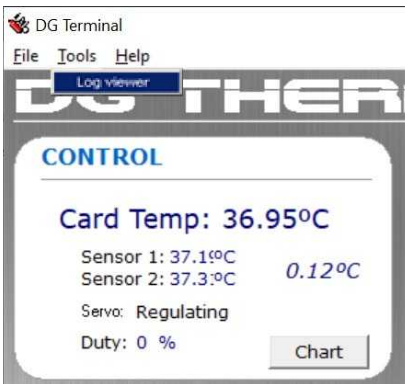


Figure 16. Log viewer command in Tools menu

If not already performed, data should be loaded from the non volatile memory.

Figure 17. Load EEPROM window

A screen with all log information will be shown with an indication of the date and time, in local format, for each event.

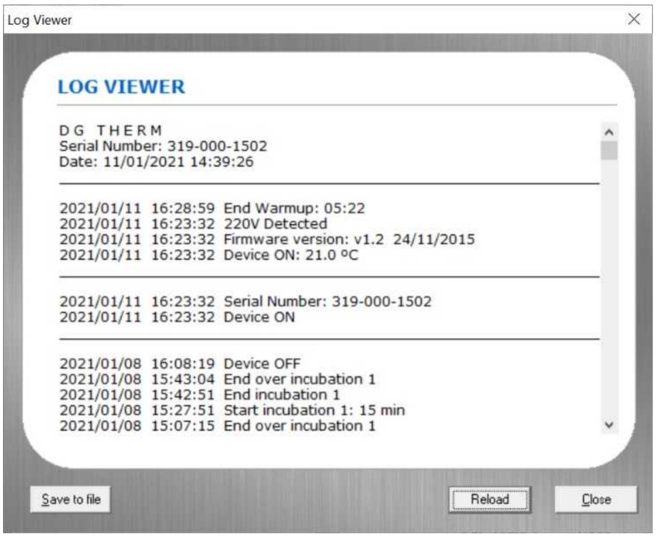


Figure 18. Log Viewer window

To save the log information into a file, use the *Save to file* button.

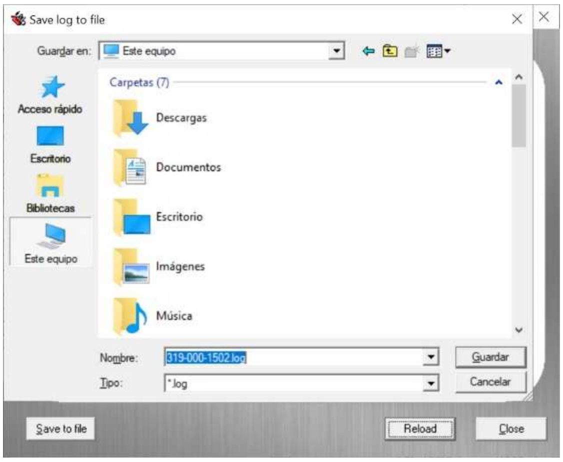


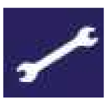
Figure 19. Save log to file window

5 Preventive Maintenance

The maintenance of the *DG Therm* incubator only includes cleaning and decontamination: See chapter 7 *"Maintenance"* in the *"DG Therm – Instructions for Use"*.

6 Corrective Maintenance

6.1 Troubleshooting

INDICATION	DESCRIPTION	POSSIBLE CAUSES	SOLUTION
-	LCD does not light up, buzzer is making noise.	Display cable is damaged or badly connected.	Replace or connect display cable.
		PCA Display does not work properly.	Replace PCA Display.
		PCA Incubator does not work properly.	Replace PCA Incubator.
-	Keypad does not work.	The keypad is badly connected or damaged.	Connect or replace keypad.
-	Display does not work and thermal static block does not heat.	A fuse has blown.	Replace PCA Incubator.
-	Cards are not lifted when opening the lids.	Trays are not present.	Place trays in the incubator.
-	A firmware update cannot be performed because communication errors have been detected.	Communication error with DG Terminal.	Check that the communications cable is well attached to both the PC and DG Therm instrument. Close DG Terminal, open it again and try to update the firmware. If the problem persists, replace the cable, RS232-USB converter (if used) or try with another PC.
E1 	Real Time Clock does not work properly.	Display cable is damaged or badly connected.	Replace or connect display cable.
		PCA Display does not work properly.	Replace PCA Display.
		PCA Incubator does not work properly.	Replace PCA Incubator.

E2	I2C communication fails with the LCD	PCA Incubator does not work properly	Replace PCA Incubator.
		Display cable is damaged or badly connected.	Replace or connect the display cable.
		PCA Display does not work properly.	Replace PCA Display.
E3	Non volatile memory does not work properly.	PCA Incubator does not work properly.	Replace PCA Incubator.
E4	Temperature sensor 1 does not work properly and I2C communication fails.	PCA Incubator does not work properly.	Replace PCA Incubator.
E5	Temperature sensor 2 does not work properly and I2C communication fails.	PCA Incubator does not work properly.	Replace PCA Incubator.
E6	Temperature sensor 1 or 2 does not work properly and the temperature difference is higher than 1.5 ºC.	PCA Incubator does not work properly.	Replace PCA Incubator.
E7	Thermostatic block does not heat (the warm up period took more than 30 minutes).	Foil heater cable is damaged or badly connected.	Connect foil heater cable or replace foil heater.
		Foil heater is damaged (check with a voltmeter if the heater impedance is between 60 Ω and 75 Ω).	Replace foil heater.
		PCA Incubator does not work properly.	Replace PCA Incubator.
E8	Thermal block temperature is higher than 50°C.	PCA Incubator does not work properly.	Replace PCA Incubator.

Table 1. Troubleshooting

After a corrective action, perform an electric continuity test.

DANGER!

The aim of this test is to determine that there is continuity between accessible dead-metal parts of the product and the grounding pin or blade of the attachment plug. If all accessible dead metal is connected, only a single test need be performed: a visual or audible device (ohmmeter, buzzer, etc) may be used to indicate grounding continuity.

Check the electric continuity between the earth terminal of the IEC inlet and the thermal block.

Figure 20. Earth terminal of the IEC inlet (left picture) and thermal block (right picture) in DG Therm

Finally execute the OQ procedure to assure that it works properly.

6.2 Procedures and Records

6.2.1 Updating the firmware (firmware version ≥ 1.3.0 and DG Terminal ≥ v1.3.0)

NOTE:
Do not operate with the instrument and do not interrupt the process while a firmware download operation is being performed.

NOTE:
DG Terminal version 1.3.0 is required for a subsequent DG Therm instrument upgrade to Firmware version ≥ 1.3.0.

The incubator *DG Therm* incorporates a firmware stored in the Texas MSP430F149 microprocessor's *flash* memory. To install a new firmware, it is needed the following:

- DG Therm instrument
- The technical service program *DG Terminal*.
- A serial cable connected to a PC port.
- A file with the version of firmware.
- A file with the kernel file.

1. Connect a power cord to the power connector located at the back of the DG Therm instrument. Plug the power cord into the mains and turn on the instrument.
2. Connect the communications cable to the instrument and to the PC where DG Terminal and the firmware and kernel files are available.
3. Open DG Terminal and in the dropdown menu of the combo box control of the Communication area, select the port which is connected to the DG Therm instrument. A new window will appear, displaying the caption Checking communications.

Figure 21. Check communications window

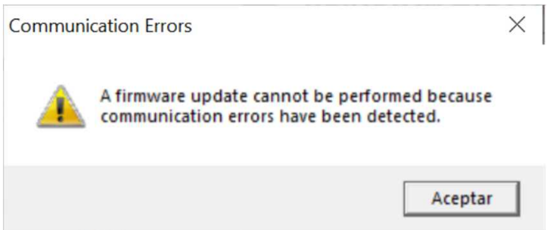
4. Wait for a few seconds until the previous window is automatically closed. Communication status will be displayed as OK in the DG Terminal main window.

Figure 22. Communication port status

5. In the Firmware area, click on Update button.
6. If the Communications Check has determined that communications are robust enough, the Update Firmware window will be displayed.

NOTE:

If there are communication errors, the following window will appear. If this is the case check *Troubleshooting* section on how to proceed.



7. In the Update Firmware window, click Open button, a file explorer window will appear, browse and select the kernel file and the new firmware version to be installed. Once both files have been selected, the Update button will be enabled. Click the Update button to start the installation of the new firmware.

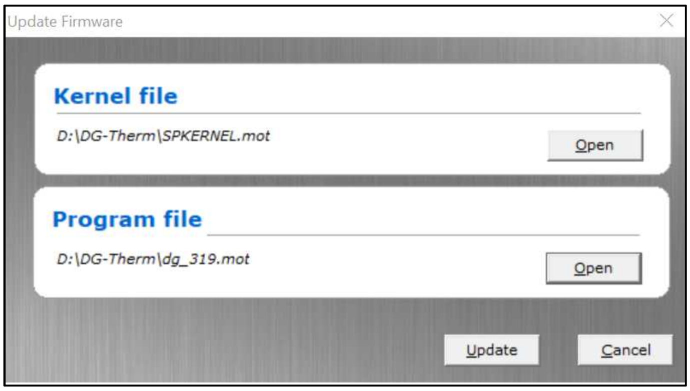


Figure 23. Update Firmware window

8. The kernel program will be installed into the microcontroller.

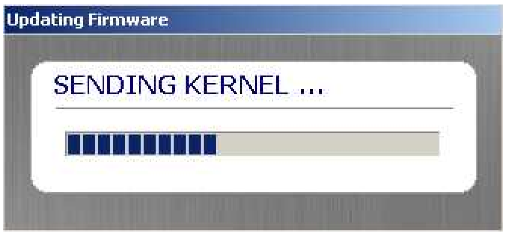


Figure 24. SENDING KERNEL window

9. After the kernel installation, it will be activated.

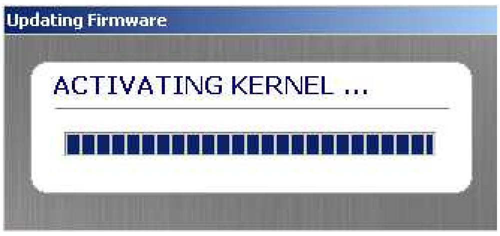


Figure 25. ACTIVATING KERNEL window

10. The new firmware will be installed into the microcontroller flash memory.

Figure 26. SENDING NEW FIRMWARE window

11. The device will be reset.

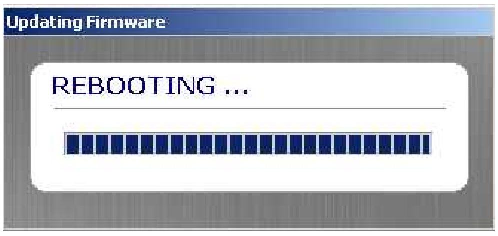


Figure 27. REBOOTING window

12. If the update process has been performed properly a confirmation will be shown on the screen. Click OK button to finish the process.

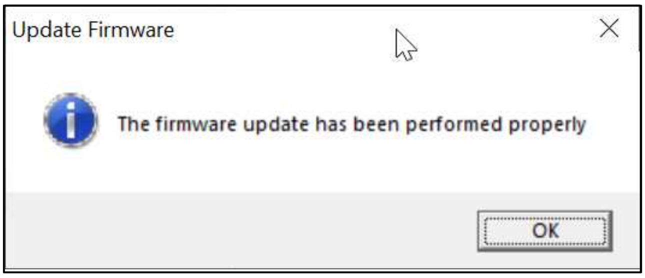


Figure 28. Update firmware process finished properly message

13. Firmware version can be checked in the Firmware area in DG Terminal.

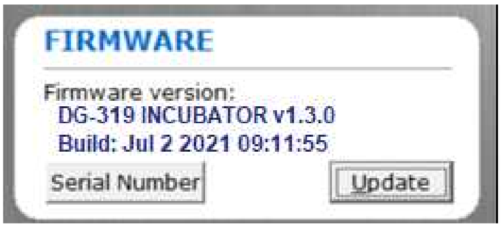


Figure 29. Firmware version on DG Terminal

14. Switch off the instrument and disconnect the communication cable.
15. Switch on the instrument, the firmware version is displayed briefly on its display screen when the instrument is turned on.

Figure 30. Firmware version on display

6.2.2 Replacement of the PCA Incubator

> **WARNING!**
> It is recommended to turn off the instrument before performing the replacement procedures.

> **WARNING!**
> To completely disconnect the apparatus from mains, remove the mains plug from the socket outlet or the appliance coupler.

1. Switch off the instrument.
2. Open the incubator using drawing S0709V0 (*DG Therm* Assembly).
3. Remove the PCA Incubator.
4. Replace the thermal silicon adhesive if it is damaged.
5. Place the new PCA Incubator.
6. Close the incubator using drawing S0709V0 (*DG Therm* Assembly).
7. Perform an electric continuity test
8. Execute the OQ procedure to assure that the incubator *DG Therm* works properly.

6.2.3 Replacement of the foil heater

> **WARNING!**
> It is recommended to turn off the instrument before performing the replacement procedures.

> **WARNING!**
> To completely disconnect the apparatus from mains, remove the mains plug from the socket outlet or the appliance coupler.

1. Switch off the instrument.

2. Open the incubator using drawing S0709V0 (*DG Therm* Assembly).
3. Remove the foil heater.
4. Remove the remaining adhesive in the heating block using alcohol.
5. Check with a voltmeter that the new heater has an impedance between 60Ω and 75Ω.
6. Place the new heater. Be sure that the heater makes contact in all the block surface and bubbles do not remain.
7. Close the incubator using drawing S0709V0 (*DG Therm* Assembly).
8. Perform an electric continuity test
9. Execute the OQ procedure to assure that the incubator *DG Therm* works properly.

6.2.4 Temperature adjustment

This procedure should be performed after a replacement of an internal part of the incubator (PCA, foil heater, heater block, etc.) and when the temperature measured with a calibrated thermometer does not correspond with the temperature displayed in the *DG Terminal* software or in the incubator display. The difference between both temperatures can be related to some losses in the heat transmission and variability of the components, characterized as ±2ºC.

Switch on the *DG Therm* and wait minimum 15 minutes to stabilize the temperature. In order to introduce the temperature read with the external thermometer, press '*Adj. Temp.'* button in COMMANDS window in the *DG Terminal* software.

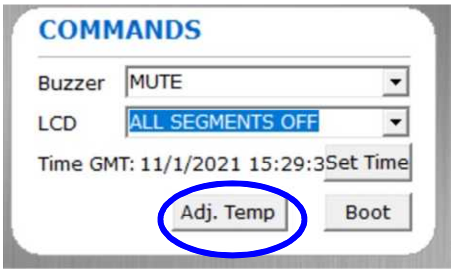


Figure 31. '*Adj. Temp.*' button in COMMANDS window in *DG Terminal* software

A new window (*Adjust Temperature*) is open and the temperature can be entered in the box.

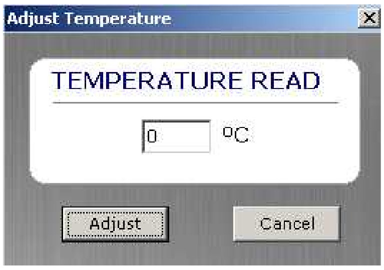


Figure 32. Adjust Temperature window

The introduced value must be between 35ºC and 39ºC. Otherwise, an error message will appear on the screen.

Figure 33. Error message related to the range of temperatures during temperature adjustment

The value introduced must have the decimals separated with a comma. Otherwise, an error message will appear on the screen.

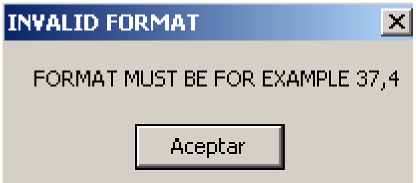


Figure 34. Error message related to the format of introduced temperature during temperature adjustment

Once the temperature is correctly introduced, you can check in the *DG Terminal* software if the measured temperature corresponds to the temperature displayed in the software.

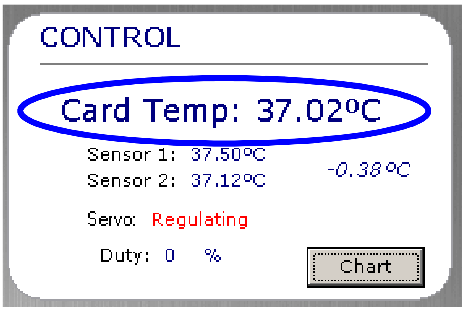


Figure 35. Card temperature in CONTROL window in *DG Terminal* software

7 Additional information

7.1 Log messages

A complete overview of all possible log events is shown in the table below:

NORMAL OPERATION	
Log message	**Explanation**
110V Detected	Device supply voltage is 110V.
220V Detected	Device supply voltage is 220V.
Abort incubation 1	First incubation has been aborted.
Abort incubation 2	Second incubation has been aborted.

Boot and reset	An external boot command has reset the device.
Change incubation time 1	First incubation time has been changed.
Change incubation time 2	Second incubation time has been changed.
Device OFF	Device switched off.
Device ON: xx.x ºC	Device switched on with indication of initial temperature.
End incubation 1	First incubation has ended.
End incubation 2	Second incubation has ended.
End overincubation 1	When user press the key 1 to reset the counter.
End overincubation 2	When user press the key 2 to reset the counter.
End Warmup: xx:xx	End of warm up period with an indication of the duration.
Firmware update	A new firmware version has been installed.
Firmware version: version and date	Actual firmware version installed.
Start incubation 1: 15 min	Start of first incubation with an indication of the programmed incubation time.
Start incubation 2: 15 min	Start of second incubation with an indication of the programmed incubation time.

Table 2. List of log messages during normal operation

WARNINGS	
Log message	**Explanation**
TEMPERATURE OUT OF MARGIN	The incubation temperature is not within the specified margins (±1ºC).
RTC OUT OF BATTERY	The Real Time Clock battery is low and it has lost the date and time stored.

Table 3. List of warning log messages

SYSTEM ERRORS	
Log message	**Explanation**
RTC ERROR (E1)	I2C communication fails with the Real Time Clock.
LCD ERROR (E2)	I2C communication fails with the LCD.

EEPROM ERROR (E3)	I2C communication fails with the EEPROM memory.
SENSOR 1 ERROR (E4)	- The temperature read out by sensor 1 is below -7°C. - The temperature read out by sensor 1 is higher than 70°C. - I2C communication fails with sensor 1.
SENSOR 2 ERROR (E5)	- The temperature read out by sensor 2 is below -7°C. - The temperature read out by sensor 2 is higher than 70°C. - I2C communication fails with sensor 2.
SENSOR DISPERSION (E6)	The difference in temperature between the temperature sensors is higher than 1,5°C.
WARMUP ERROR (E7)	The warm up period took more than 30 minutes.
OVERTEMPERATURE (E8)	The thermal block temperature is higher than 50°C.

Table 4. List of system errors log messages

CONFIGURATION MODIFICATION	
Log message	**Explanation**
Configuration change: Buzzer Volume	Buzzer volume changed.
Configuration change: Temperature adjusted	Temperature adjustment has been performed.
Configuration change: Time	Time and date changed.
Configuration change: Units	Units changed between °C and ºF.

Table 5. List of configuration modification log messages

8 Drawings

This chapter describes the available options when viewing the digital drawings representing all the modules of DG Therm instrument, including 3D views and animated assembly procedures that are performed in factory for each of them. As detailed below, a free software is required to view the drawings and assembly procedures, which are contained in '.pvz' files as an attachment to the current Service Manual document.

8.1 Requirements

- 3D drawings ('.pvz' files)

The file name format of the 3D drawings is *MxxxxxxxVn_myy.pvz*, where *xxxxxxx* is the module identification number, *n* is a number indicating the drawing version, and *yy* is the modification number for the corresponding module. 3D drawings are included in the *Drawings* folder as part of the current Service Manual.

- Drawings table

This table can be used to find the name of the drawing corresponding to each module.

- Spare parts list

The raw material codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.

- 3D viewer

Creo View Express is the software used to view the available 3D drawings and the corresponding assembly procedures. It can be downloaded following the instructions provided in section *8.3.1 Installation* and configured as detailed in section *8.3.2 Configuration*.

8.2 Drawings reference table

There are the following 3D drawings (pvz format):

Module	Description	Number
GEN	General assembly	M6007524V0_m00.pvz

Table 6. *List of 3D drawings*

Additionally, there are the following mechanical drawings (2D, pdf format):

Module	Description	Number
PACK	Packaging	S1247v0_m02.pdf
GEN	General assembly	S0709v0_m11.pdf
	De-assembly	S0798v0_m01.pdf

Table 7*. List of 2D drawings*

8.3 Creo View Express

8.3.1 Installation

- **Within Grifols company**

  Install **Creo View Express** from Grifols Software Center.

- **External distributors**

  Look for Creo View Express in https://www.ptc.com (free viewer) and follow the downloading and installation instructions.

NOTE:

Use Creo View Express software version 8.0 (or higher).

8.3.2 Configuration

Once a drawing ('.pvz' file) is opened using Creo View Express, the main structure of that module can be displayed by selecting the *Structure* tab on the left panel of the screen, then double-clicking on the main structure code contained in the *Ilustración* field.

NOTE:

Some drawings may contain more than one main structure, so they should be opened individually.

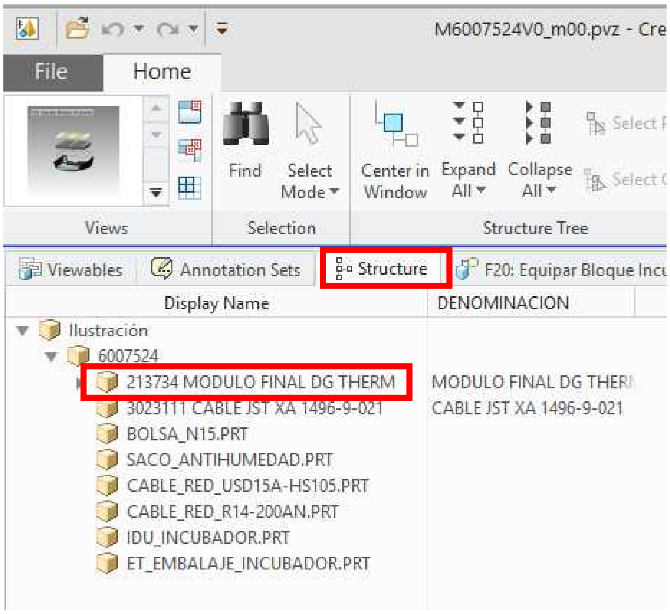


***Figure 36.** Main structure code*

A new tab named *Model* will be created, and the 3D drawing of the selected module will be shown on screen. At this point, several drawing techniques can be used to represent the model by selecting the *Render Mode* button in Creo View Express toolbar (the shaded view is selected by default).

When selecting any of the parts composing the drawing, the corresponding component will be highlighted in the navigation tree on the left side of the screen. In order to see the raw material description and its code (factory reference) instead of the default values, the following configuration should be applied:

NOTE:

This configuration is required if raw material codes need to be searched within the module

1. Go to *File/Options*.
2. Go to *General/Navigation* window.
3. Enable the *Use alternate path names* checkbox and click on *Configure*. A new window *Configure Alternate Part Names* opens.
4. Click on *Insert Attribute…*.
5. Select 'CODIGO' and click on *Insert value*.
6. Select 'DENOMINACION' and click on *Insert value*.
7. Consider applying a whitespace between both values in the *Configure Alternate Part Names* window.

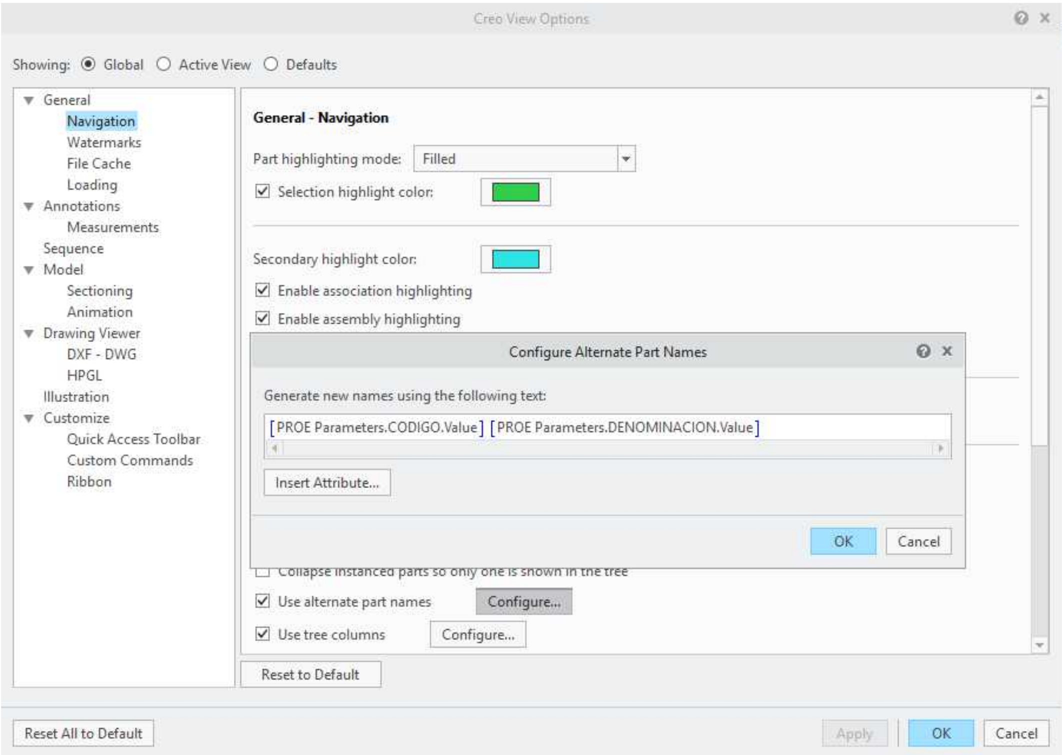


Figure 37. *Creo View Options*

8. Click on *Close*, then *OK*.
9. Click on *Apply*, then *OK*.
10. Now the raw material description and its factory reference can be seen in the navigation tree for each part composing the drawing. This reference can also be found when searching for *CODIGO* field.

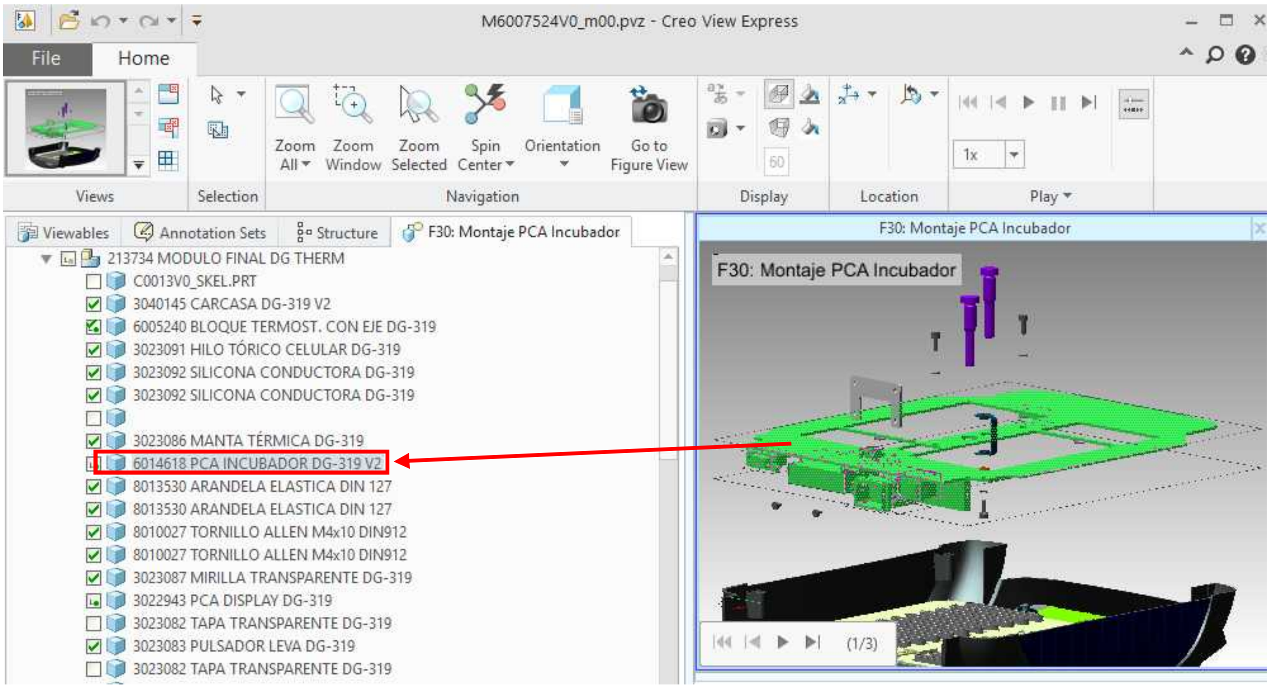


Figure 38. *Highlighted part description after selecting a drawing part*

8.3.3 Models structure and navigation

8.3.3.1 Tree structure

1. Click on *Structure* tab and double click on the main structure code contained in the *Ilustración* field.

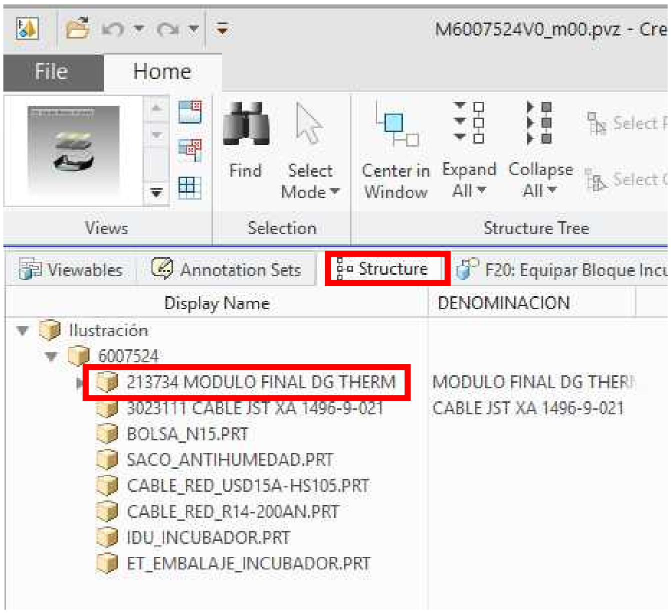


Figure 39. *Main structure code*

2. A new tab named *Model* will be created, and the main model of the selected module will be shown on screen. The tree structure (starting on the main structure code) can be expanded or collapsed by clicking on the triangles next to each field.

3. When double clicking in any field of the tree, the corresponding component in the drawing is zoomed in and changes its colour to blue (subassembly containing raw materials) or green (raw materials).

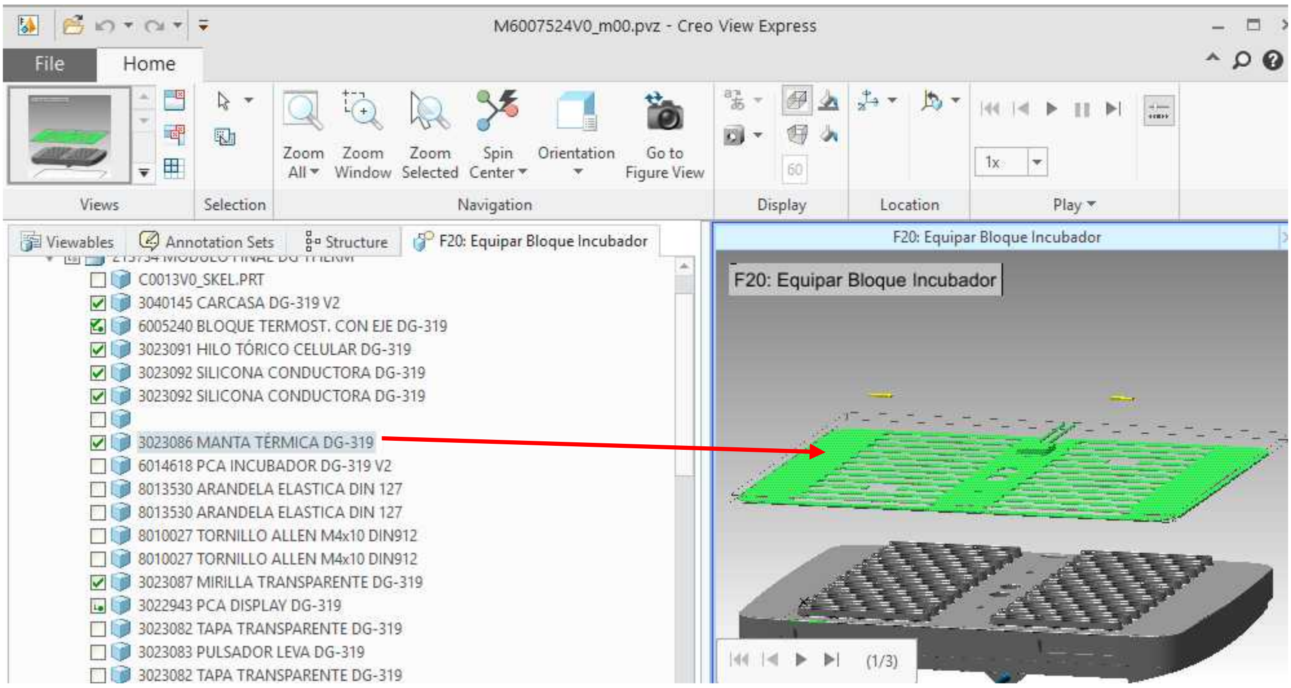


Figure 40. *Tree structure component selection*

4. Parts can be hidden or isolated by right-clicking on them and selecting the desired action, and then reestablished by right-clicking anywhere within the drawing area and selecting *Unhide All*. They can also be hidden or unhidden by enabling/disabling the check square near each component in the tree structure.

8.3.3.2 Models navigation

Once the drawing is displayed on screen, it can be rotated, dragged, zoomed in or out, etc. by using the below listed functionalities for Creo View Express:

- Zoom in/out (pointer area taken as reference point):
  - Mouse: rotate the mouse wheel.
  - Pad: put two fingers on pad and move them up or down.

- Rotate model:
  - Mouse: Press and hold right button and drag mouse.
  - Pad:
    - Select *Free Rotation* button in Creo View Express toolbar.
    - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
    - Hold left click on the sphere while dragging on the pad.

- Drag model:
  - Mouse: Press the wheel and drag mouse.
  - Pad:
    - Select *Transform* button in Creo View Express toolbar.
    - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
    - Hold left click on a sphere's center grey circle while dragging on the pad.

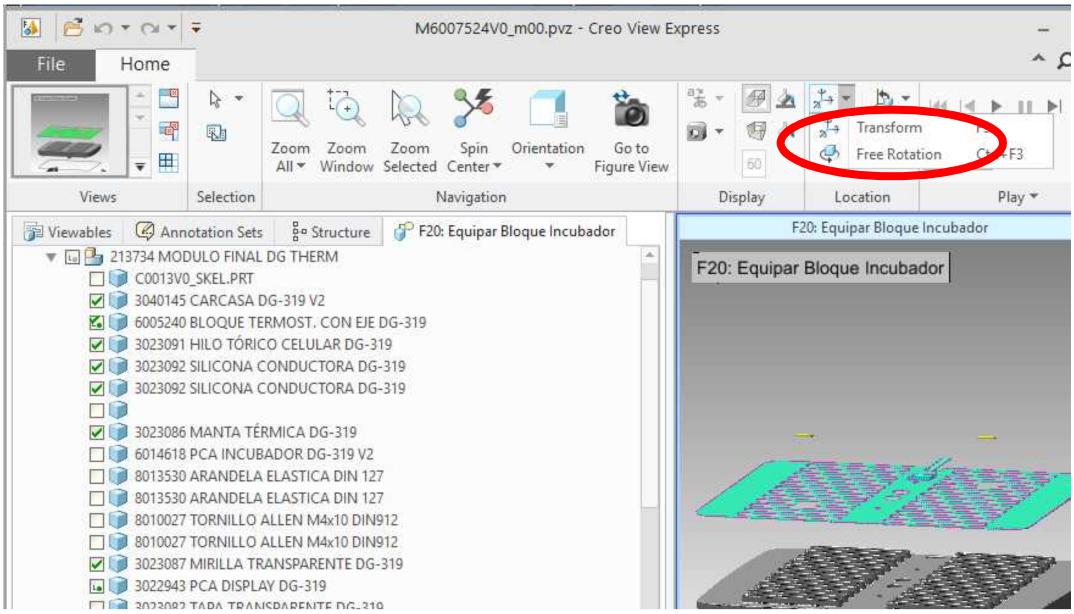


Figure 41. *Transform options in Creo View Express toolbar*

- Hide/Unhide part:
  - Select the part clicking by right-clicking on it and select *Hide* or *Unhide All*.
  - Select the part clicking on it and enable or disable it in the tree structure.

- Center model on screen: double click on the background

- Restore location: click on *Restore location* button in Creo View Express toolbar. With this, any part that has been moved/rotated from its default position within the drawing area will restore its original position.

8.3.3.3 Animated assembly procedures

The *Viewables* tab contains a list of animated assembly procedures detailing the assembly steps as done in factory for each of the modules composing DG Therm instrument. Each component being assembled in an animation is identified with its raw material code.

1. When opening a '.pvz' file for a specific module, the *Viewables* tab is selected by default, showing the corresponding animated assembly procedures (*Figures* area).

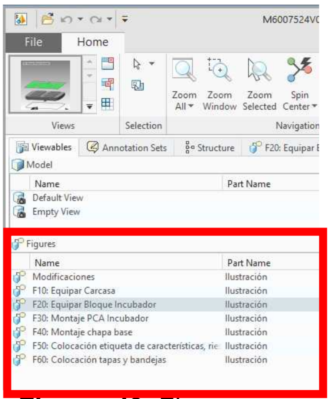


Figure 42. *Figures area*

2. Double click on the desired assembly and a new tab will be created, showing the tree structure with the parts involved along with the corresponding animated assembly.

3. The animation can be played and paused for each of the steps composing it by using the *Playback* control buttons at the bottom left corner of the animation screen, or those at the *Play* section in Creo View Express toolbar.

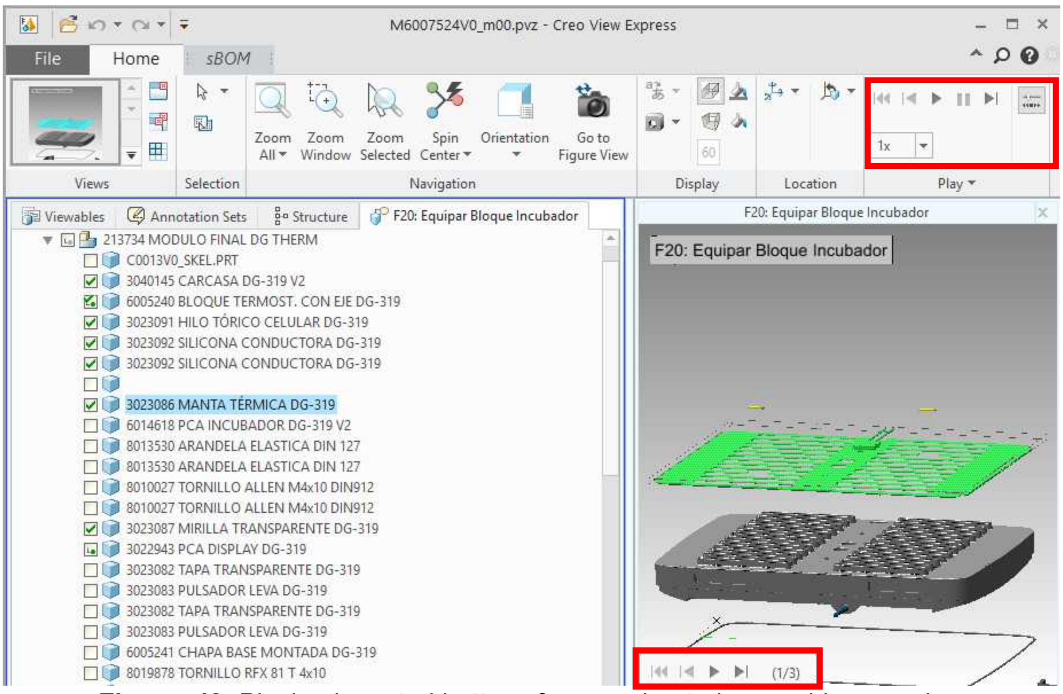


Figure 43. *Playback control buttons for an animated assembly procedure*

4. Parts can be hidden/unhidden during the animation by enabling/disabling the square near each component in the tree structure or selecting the corresponding option after right-clicking on it.

5. The background colours can also be modified as desired by editing the *Background* and *Gradient* options of the *Display* section in Creo View Express toolbar.

8.3.3.4 Spare parts searching

The raw codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.

> **NOTE:**
> Before searching for a specific raw material code, make sure that the configuration detailed in section *8.3.2 Configuration* has been applied

8.3.3.5 Finding a drawing part for a spare parts list's reference code

1. Copy the desired drawing reference code from the Spare Parts List 'Drawing reference' column.
2. Search in the drawings table the drawing of the module where the part is assembled.
3. Click on *Find* button in Creo View Express toolbar.
4. Paste the previously copied code in the new window and click on *Find next*.
5. The raw material corresponding to the selected drawing reference will be highlighted both in the tree structure and in the model drawing.

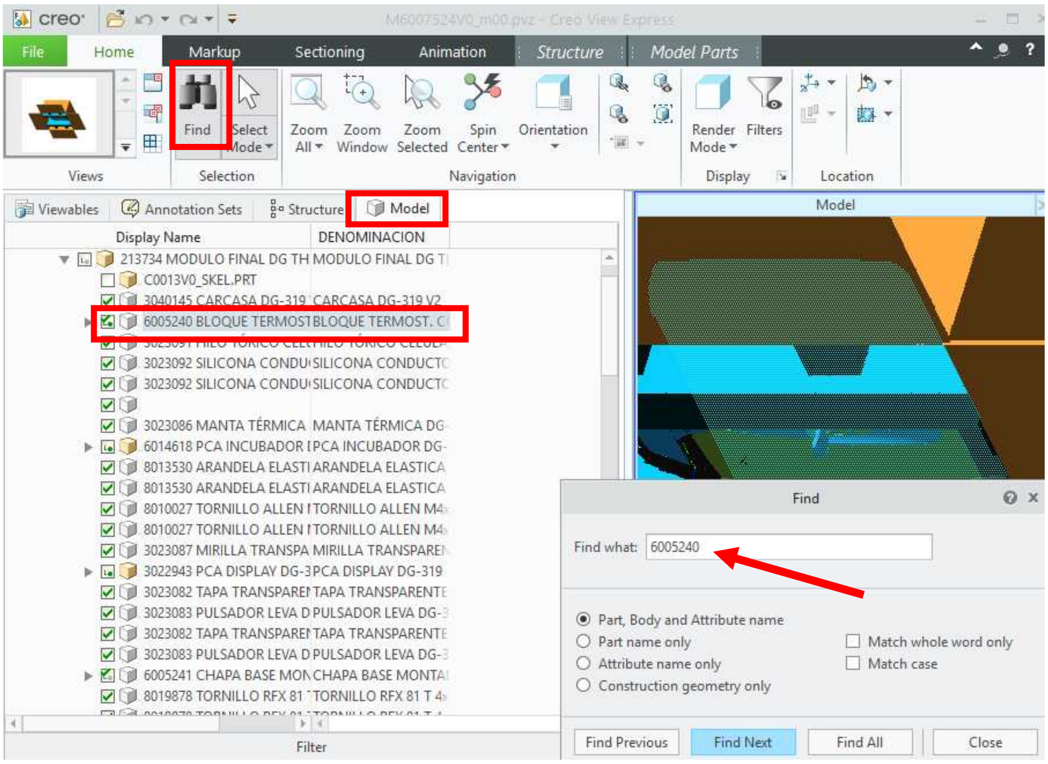


Figure 44. *Finding a drawing part from a known code*

8.3.3.6 Finding the spare part list's reference code for a drawing part

1. Select the part of the drawing which raw material code needs to be known.
2. The part will be selected in the tree structure. If the view configuration detailed in section *8.3.2 Configuration* 8.3.2has been applied, the raw material code (*CODIGO* field) will be listed as part of the

material description. The raw material code (*CODIGO* field) can also be seen by enabling the lower data panel option on the lower left side of the screen.

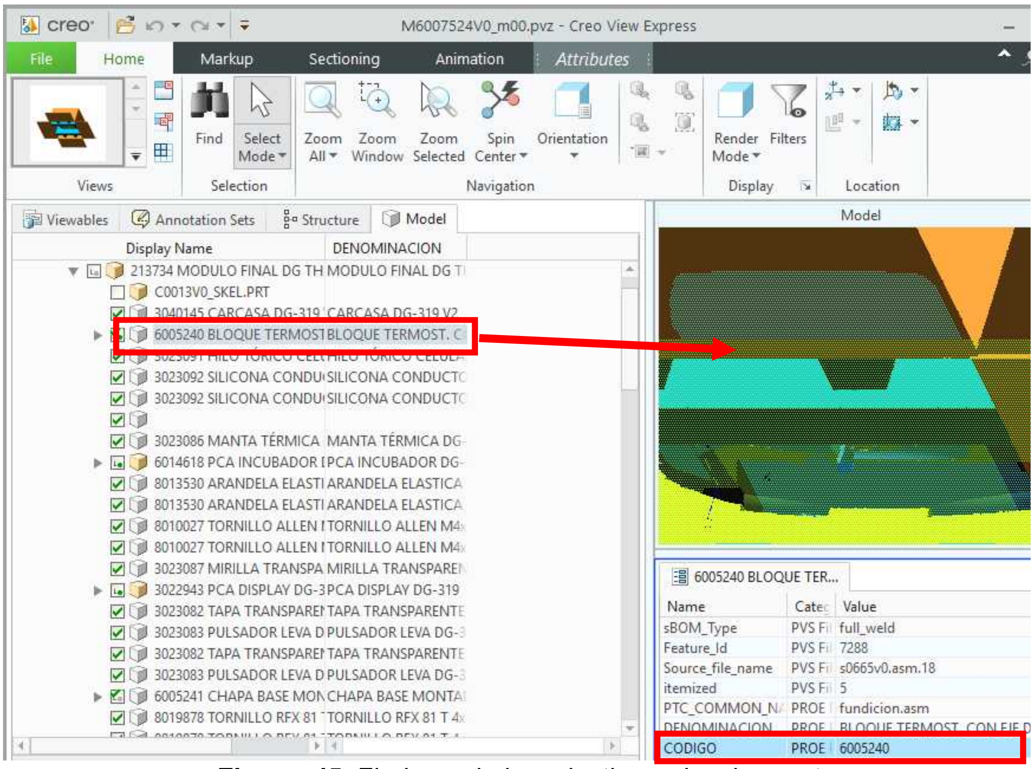


Figure 45. *Find a code by selecting a drawing part*

3. Search for the obtained raw material code in DG Therm Spare Parts List. The code will be listed in the *Drawing Reference* column for one or more spare part codes, so the desired reference can be obtained. If the raw material code cannot be found in the Spare Parts list:

   a) If the raw material code is part of a set of codes included in a 'parent' code on top of it in the tree structure, try to look for the parent code in the Spare Parts list. The parent code of a specific part can be automatically selected by selecting it and then clicking on the *Select Parent* button than can be found in the *Structure* tab in Creo View Express toolbar.

   b) If the raw material code is not part of a set of codes included in a 'parent' code, or if the parent code cannot be found in the Spare Parts list, it means that no part number reference has been assigned to the desired raw material code.